CS & IT ENGINEERING

Operating System

CPU Scheduling

Lecture -4

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Recap of Previous Lecture

doubt ?

Topic SRTF Scheduling

Topic LJF Scheduling

Topic LRTF Scheduling

Topics to be Covered

Topic Priority based Scheduling

Topic Round Robin Scheduling

Topic : HRRN (Highest Response Ratio Next)

Process	Arrival Time	Burst Time
P1	0	6
P2	1	4
P3	5	1
P4	6	5

P1	P2	P3	P4

0 6 10 11 16

At time 6:-

$$RR(P2) = \frac{5+4}{4} = 2.25 \text{ (Highest)}$$

$$RR(P3) = \frac{1+1}{1} = 2$$

$$RR(P4) = \frac{0+5}{5} = 1$$

At time 10:-

$$RR(P3) = \frac{5+1}{1} = 6 \text{ (Highest)}$$

$$RR(P4) = \frac{4+5}{5} = 2.2$$

Topic : Priority Based Algorithm

Scheduling Criteria: Highest priority process first | Tie breaker ⇒ given in questⁿ

Type of Algorithm: Non-preemptive
or
preemptive

Topic : Priority Based Algorithm

Process	Arrival Time	Burst Time	Priority
P1	0	4	4
P2	1	2	5
P3	2	3	6
P4	3	1	10(Highest)
P5	4	2	9

Non-preemptive :-

P1	P4	P5	P3	P2

0 4 5 7 10 12

Topic : Priority Based Algorithm

Process	Arrival Time	Burst Time	Priority
P1	0	4	4
P2	1	2	5
P3	2	3	6
P4	3	1	10(Highest)
P5	4	2	9

Preemptive :-

P1	P2	P3	P4	P5	P3	P2	P1

0 1 2 3 4 6 8 9 12

Topic : Priority Based Algorithm Question Non-Preemptive

Process	Arrival Time	Burst Time	Priority
P1	0	7	9
P2	1	3	4
P3	2	5	2
P4	3	2	1 (Highest)
P5	4	6	3
P6	5	1	8

P1	P4	P3	P5	P2	P6

0 7 9 14 20 23 24

Topic : Priority Based Algorithm Question Preemptive

Process	Arrival Time	Burst Time	Priority
P1	0	7	9
P2	1	3	4
P3	2	5	2
P4	3	2	1 (Highest)
P5	4	6	3
P6	5	1	8

P1	P2	P3	P4	P3	P5	P2	P6	P1

0 1 2 3 5 9 15 17 18 24

Ques) Preemptive

Process	AT	BT	Priority
P1	0	6	4
P2	1	8	5
P3	8	4	7 (Highest)
P4	2	5	6

P1	P2	P4	P2	P3	P2	P1

0 1 2 7 8 12 18 23

Topic : Priority Based Algorithm

Advantages:

1. Better response for real time situations

Disadvantages:

2. Low Priority Processes may suffer from starvation

Priority

static

same priority for a process and it won't change.

dynamic

priority of processes may change

solution of starvation $\Rightarrow$ Aging

increase priority of process when it waits for a certain time.

Topic : Round Robin (RR)

Scheduling Criteria: Arrival time + Quantum time

Type of Algorithm: Preemptive

Topic : Round Robin (RR)

Q = 2

Process	Arrival Time	Burst Time
P1	0	3
P2	0	6
P3	0	4
P4	0	5

Ready Queue

P1, P2, P3, P4

P1	P2	P3	P4	P1	P2	P3	P4	P2	P4

0 2 4 6 8 9 11 13 15 17 18

Topic : Round Robin (RR)

$Q = 2$

Process	Arrival Time	Burst Time
P1	0	5
P2	1	4
P3	2	1
P4	3	3

Time	Ready Queue
0	~~P1~~
2	~~P2~~, P3, P1
4	P3, P1, P4, P2

P1	P2	P3	P1	P4	P2	P1	P4

0 2 4 5 7 9 11 12 13

Topic : Round Robin (RR)

$Q = 2$

Process	Arrival Time	Burst Time
P1	0	6
P2	1	5
P3	2	4
P4	3	3
P5	4	2
P6	5	4

Time	Ready Queue
0	~~P1~~
2	~~P2~~, P3, P1
4	~~P3~~, P1, P4, P5, P2
6	~~P1~~, ~~P4~~, ~~P5~~, ~~P2~~, ~~P6~~, ~~P3~~

no. of context switches
$= \underline{1}2$

P1	P2	P3	P1	P4	P5	P2	P6	P3	P1	P4	P2	P6

0 2 4 6 8 10 12 14 16 18 20 21 22 24

Topic : Round Robin (RR)

Q = 3

Process	Arrival Time	Burst Time
P1	5	~~6~~ ~~3~~
P2	8	~~7~~ ~~4~~ 1
P3	3	~~8~~ ~~5~~ ~~2~~
P4	6	~~3~~
P5	2	~~2~~ 0
P6	4	~~4~~ ~~1~~

Time	Ready Queue
0	
2	~~P5~~
4	~~P3~~, P6
7	~~P6~~, P1, P4, P3
10	P1, P4, P3, P2, P6

	P5	P3	P6	P1	P4	P3	P2	P6	P1	P3	P2	P2
0–2	2–4	4–7	7–10	10–13	13–16	16–19	19–22	22–23	23–26	26–28	28–31	31–32

0 2 4 7 10 13 16 19 22 23 26 28 31 32

Topic : Round Robin (RR)

H.W.

Q = 3

Process	Arrival Time	Burst Time
P1	0	12
P2	0	5
P3	3	9
P4	5	6
P5	2	8
P6	4	2
P7	1	7

2 mins Summary

Topic Priority based Scheduling

Topic Round Robin Scheduling

Happy Learning

THANK - YOU